Practical work	Students will not need to take any documents into the laboratory for the practical aspect of this assessment although they may bring a scientific calculator. Teachers should issue students with the (unmarked) plan of the practical that they are to carry out. If necessary, teachers may allow students to analyse results under supervision in the next lesson. In this situation, teachers must collect in the work produced by their students. Teachers should not mark the practical work. In the following lesson, the documents should be returned to students under supervised conditions. Students should not have access to any other sources of information while they are completing the analysis of their results. Teachers who opt for internal assessment should mark the practical work after students have completed the analysis of their results.
Assessing work	The assessment criteria are shown in the next section. Each criterion scores a maximum of 1 mark. The criteria are not hierarchical. The assessment of planning, recording and analysis is based on written evidence in the form of a report.
For centres marking the written report	The written evidence should be annotated. This can be done by writing the skill code eg A15 near to the appropriate section of the report and ticking the box A15 on the grid below. The marks given for the report should be submitted to Edexcel on the mark sheets as shown in the Edexcel *Coursework Guide* that will be available on the Edexcel website www.edexcel.org.uk.
For centres not marking the written report	The written report should be submitted to Edexcel as discussed in the Edexcel *Coursework Guide* that will be available on the Edexcel website www.edexcel.org.uk.
Supervision	Students must work on their own for each part of this assessment. All aspects of this assessment must be done under supervised conditions.

Assistance for students	Students may require assistance whereby the teacher needs to tell the student what they have to do. Assistance in this respect carries a penalty. The teacher should record details of any assistance provided on the student's work.
Important reminder	Students should submit their work for assessment **once only**. **Neither** the plan **nor** the experiment should be given back to students to be improved.

6.3 Assessment criteria

A: Planning

Ref	Criterion	Mark
P1	Identifies the most appropriate apparatus required for the practical in advance	1
P2	Provides clear details of apparatus required including approximate dimensions and/or component values (for example, dimensions of items such as card or string, value of resistor)	1
P3	Draws an appropriately labelled diagram of the apparatus to be used	1
P4	States how to measure one quantity using the most appropriate instrument	1
P5	Explains the choice of the measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken	1
P6	States how to measure a second quantity using the most appropriate instrument	1
P7	Explains the choice of the second measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken	1
P8	Demonstrates knowledge of correct measuring techniques	1
P9	Identifies and states how to control **all** other **relevant** quantities to make it a fair test	1
P10	Comments on whether repeat readings are appropriate for this experiment	1
P11	Comments on **all** relevant safety aspects of the experiment	1
P12	Discusses how the data collected will be used	1
P13	Identifies the main sources of uncertainty and/or systematic error	1
P14	Plan contains few grammatical or spelling errors	1
P15	Plan is structured using appropriate subheadings	1
P16	Plan is clear on first reading	1
	Maximum marks for this section	**16**

B: Implementation and measurements

Ref	Criterion	Mark
M1	Records all measurements with appropriate precision, using a table where appropriate	1
M2	Readings show appreciation of uncertainty	1
M3	Uses correct units throughout	1
M4	Refers to initial plan while working and modifies if appropriate	1
M5	Obtains an appropriate number of measurements	1
M6	Obtains measurements over an appropriate range	1
	Maximum marks for this section	**6**

C: Analysis

Ref	Criterion	Mark
A1	Produces a graph with appropriate axes (including units)	1
A2	Produces a graph using appropriate scales	1
A3	Plots points accurately	1
A4	Draws line of best fit (either a straight line or a smooth curve)	1
A5	Derives relation between two variables or determines constant	1
A6	Processes and displays data appropriately to obtain a straight line where possible, for example, using a log/log graph	1
A7	Determines gradient using large triangle	1
A8	Uses gradient with correct units	1
A9	Uses appropriate number of significant figures throughout	1
A10	Uses relevant physics principles correctly	1
A11	Uses the terms *precision* and either *accuracy* or *sensitivity* appropriately	1
A12	Discusses more than one source of error qualitatively	1
A13	Calculates errors quantitatively	1
A14	Compounds errors correctly	1
A15	Discusses realistic modifications to reduce error/improve experiment	1
A16	States a valid conclusion clearly	1
A17	Discusses final conclusion in relation to original aim of experiment	1
A18	Suggests relevant further work	1
	Maximum marks for this section	**18**

CONTEXT-LED APPROACH BASED ON THE SALTERS HORNERS ADVANCED PHYSICS PROJECT

The following section shows how the specification may be taught using the context-led approach.